

Johnson To

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. Chemical Engineering

Expected Graduation – May 2028

- **Coursework:** Physics for Scientists and Engineers, Chemical Process Analysis, Math & Statistics in ChemE, Organic Chemistry, Chemical Engineering Thermodynamics, Transport Processes, Linear Algebra & ODEs
- **Activities:** American Institute of Chemical Engineers

PROFESSIONAL EXPERIENCE

Research & Development Engineering Intern

Hayward, CA

Delphon Industries

June 2026 – August 2026

- Optimized high-temperature silicone ESD films by tuning cure concentration to reach $10^7 \Omega/\text{sq}$ surface resistivity
- Fabricated 50+ thin-film drawdowns using automated and manual applicators, vacuum degassing, and oven curing
- Characterized thin films using SEM, Zygo profilometry, optical microscopy, load cells, and multimeter measurements
- Tracked tack force of 5 CNT formulations in 4-week aging, confirming stability at 150°C ; presented results to R&D
- Built an AI-assisted Python workflow to parse Zygo .datx (HDF5) files and apply Otsu thresholding to 40 samples

Private Tutor

Bakersfield, CA

Self-Employed

December 2022 – May 2024

- Delivered 400+ hours of personalized algebra and trigonometry instruction using structured problem-solving methods
- Developed 50+ customized lesson plans using student performance data to target weaknesses and reinforce concepts

PROJECT EXPERIENCE

Hydrogen Electrolysis Optimization Project

Berkeley, CA

Personal Project

January 2026 – May 2026

- Built an electrolysis cell with graphite electrodes and NaHCO_3 electrolyte in distilled water, operating below 30 V
- Collected 50+ trials of voltage, current, and gas output data to model hydrogen yield versus voltage and current
- Compared Faraday's law predictions with measured H_2 output, reaching a peak Faradaic efficiency of about 90%
- Performed experimental error analysis to identify systematic inefficiencies and validate reproducibility across trials

Ibuprofen Concentration and Degradation Analysis

Berkeley, CA

Chemistry Lab Project

January 2026 – May 2026

- Quantified ibuprofen concentration in commercial tablets via acid-base titration using 1:1 NaOH neutralization
- Constructed a 4-point calibration curve in Python (matplotlib) correlating NaOH volume to ibuprofen concentration
- Conducted 10+ titration trials across UV-radiated, heat-exposed, and control groups to assess degradation effects

Smart Waste Sorting System

Berkeley, CA

Personal Project

January 2026 – May 2026

- Machined wood and polycarbonate components to construct a $15'' \times 30'' \times 20''$ enclosure with hinge mechanism
- Deployed a ResNet50 CNN via Raspberry Pi and Flask, reaching up to 95% confidence classifying trash vs. recycling
- Integrated camera module, servo actuation, and confidence-threshold logic to enable real-time automated decisions

LEADERSHIP & ACTIVITIES

Media Committee Member

Berkeley, CA

American Institute of Chemical Engineers (AIChE)

August 2025 – December 2025

- Created 10+ graphics w/ marketing materials to promote professional development, networking, & community events
- Contributed to weekly media planning meetings to coordinate content strategy and improve member outreach
- Designed digital content for social media platforms, helping communicate and increase awareness of chapter activities

SKILLS & INTERESTS

Lab: SEM, Zygo Profilometry (Mx), Optical Microscopy, Thin-Film Drawdowns, Resistivity, Titration, Machining

Software: Python (matplotlib, Flask, NumPy), Excel

Interests: Golfing, Rock Climbing, Hiking & Backpacking, Collecting Vinyls